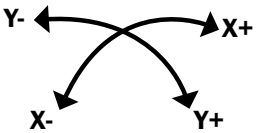
Jewell Instruments

MAKING SENSE OUT OF MOTION

eDXI-100/200

Single or Dual Axis Digital Inclinometer

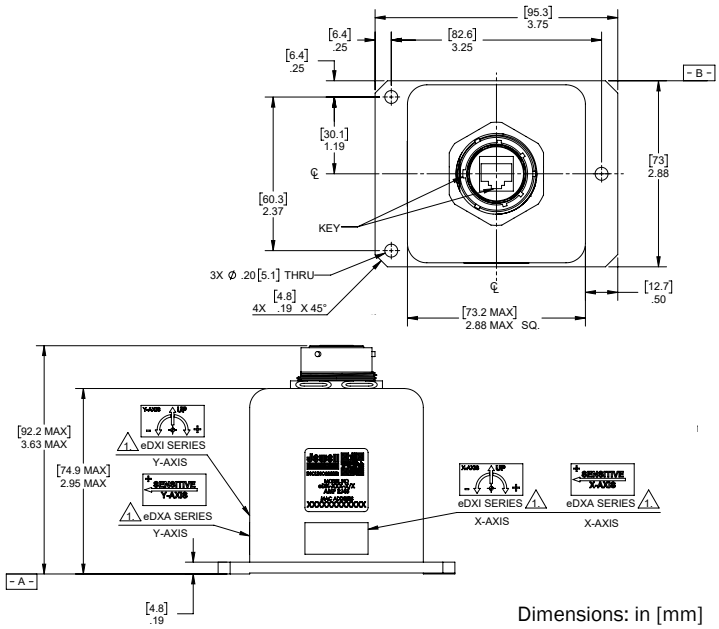
The Jewell eDXI-100/200 Series is a single or dual digital inclinometer with the convenience of Ethernet output. With its power over ethernet capability, no power source is needed. Simply plug it in to your computer.



*Standard eDXI-100 includes X-axis only

FEATURES:

- Digital output
- Resolution 0.001°
- Mechanical Shock 1500g 1msec 1/2 sine
- Industry Standard Ethernet 10 Base T or 100 Base TX (Auto-sensing)
- For use in high shock and vibration environments
- PoE (power over Ethernet)
- Low Noise



Dimensions: in [mm]

APPLICATIONS:

- Radar/Antenna Control
- Structural Monitoring
- Linear Acceleration/Deceleration Measuring
- Automatic Train Control (ATC, ATP)

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PERFORMANCE SPECIFICATIONS eDXI-100/200

PERFORMANCE

INPUT RANGE (°)	±1.0	±3.0	±14.5	±30.0	±60.0
NON LINEARITY (%FRO max)[2]	0.02	0.015	0.02	0.02	0.03
SCALE FACTOR TOLERANCE (% max)	0.1				
SCALE FACTOR TEMP. SENSITIVITY (PPM/°C, max)	100				
OUTPUT AT 0° TILT (max)	±0.01	±0.01	±0.05	±0.05	±0.05
0° OUTPUT TEMP. SENSITIVITY (°/°C, max)	0.005				
BANDWIDTH (-3dB Hz, nominal) [3]	3	6	30	30	30
TRANSVERSE AXIS MISALIGNMENT (° max)	0.2	0.2	0.5	0.5	0.5
RESOLUTION & THRESHOLD (° max)	0.001				
POWER ON REPEATABILITY (° max)	0.001				
HYSTERESIS (° max)	0.001				
REPEATABILITY (° max)	0.001	0.001	0.002	0.002	0.003

ELECTRICAL

NUMBER OF AXES	1 or 2
SUPPLY VOLTAGE RANGE (VDC)	36 to 57
START-UP VOLTAGE (VDC, min)	42
INPUT CURRENT (mA, max)	250
INPUT TO OUTPUT ISOLATION (Vpeak Impulse)	1500
NOISE (milli-degrees rms, max)	2

ENVIRONMENTAL

OPERATING TEMP. RANGE	-40° to +85°C
STORAGE TEMP. RANGE	-40° to +85°C
VIBRATION	20 grms
SHOCK	1500g, 1 msec, ½ sine

PIN OUTS

1	TX+
2	TX-
3	RX+
4	PoE45+

5	PoE45-
6	RX-
7	PoE78+
8	PoE78-

NOTES:

1. Full range is defined as “from negative full input angle to positive full input angle”
2. Non-linearity is specified as deviation of output referenced to a best fit straight line, independent of misalignment.
3. In default condition without averaging enabled.

*Specifications subject to change without notice on account of continued product development

PERFORMANCE SPECIFICATIONS eDXI-100/200

ENCLOSURE

HOUSING MATERIAL	Anodized & Alodine Aluminum
WEIGHT	eDXI-100 345 g, max
	eDXI-200 390 g, max
PROTECTION CLASS (per IEC 529)	IP65
NEMA ENCLOSURE RATING	6
SEAL	MIL-STD-202 Method 112
CONNECTOR TYPE	Amphenol RJF72B00
RECOMMENDED MATING CONN.	Amphenol RJF6B

DIGITAL OUTPUT

INTERFACE	Ethernet 10Base-T or 100Base-TX (Auto-sensing)
PROTOCOL	TCP/IP, UPD/IP, ARP, Telnet, ICMP, SNMP, DHCP, BOOTP, TFTP, AutoIP, and HTTP
OUTPUT REP.	Degrees

ORDERING INFORMATION

HOW TO ORDER

	AXES	MODEL #	PART #
±1.0	1	eDXI-100-1	02550380-101
	2	eDXI-200-1/1	02550380-201
±3.0	1	eDXI-100-3	02550380-102
	2	eDXI-200-3/3	02550380-202
±14.5	1	eDXI-100-14.5	02550380-103
	2	eDXI-200-14.5/14.5	02550380-203
±30.0	1	eDXI-100-30	02550380-104
	2	eDXI-200-30/30	02550380-204
±60.0	1	eDXI-100-60	02550380-105
	2	eDXI-200-60/60	02550380-205

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ISO9001 CERTIFIED